



FREE GUIDE · FOR COACHES WHO WANT TO PROGRAM PROPERLY

How to Program: A Framework for Building Client Training

The evidence behind progressive overload, volume and periodization, how to actually individualize it client by client, and what the research really says about training men and women differently.

Who this guide is for

This is written for coaches who already know the basics from their Certificate III/IV, but want the actual reasoning behind programming decisions, not just a template to copy. It covers the evidence base, where that evidence is strong and where it's genuinely mixed, and how to apply it to a real client in front of you rather than an average from a research paper.

Two things run through the whole guide. First, the textbook principles are a floor, not a finished program: two clients with identical goals can need completely different structures. Second, being evidence-based cuts both ways, it means using what's well supported, and being honest when a popular claim is weaker than it sounds. You'll see both used here.

A NOTE ON ACCURACY

Exercise science is an active research field, and not every finding in this guide carries the same weight. Where the evidence is strong and consistent, it's stated plainly. Where it's mixed, debated, or still emerging, that's flagged explicitly rather than smoothed over.

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What Programming Actually Is

A workout is one session. A program is a direction across weeks, where each session exists partly to set up the next one. The difference matters because a client can train hard in every single session and still not have a program, if nothing about the plan changes on purpose over time.

The loop every good program runs on

- ✓ **Assess:** current ability, goals, training history, injuries and constraints, before writing a single session.
- ✓ **Plan:** structure sessions and weeks so they build toward the goal, not just fill time.
- ✓ **Execute:** coach the session as written, but coach the person, not the paper.
- ✓ **Monitor:** track what actually happened, loads, reps, how hard it felt, how they recovered.
- ✓ **Adjust:** change the plan based on that real data, not on what looked good three weeks ago.

Why effort alone isn't enough

New clients tend to improve for months on almost any reasonable stimulus, a well-documented effect of being new to structured training (often called the novice effect). It's real, and it can hide a weak program for a while. Once that early adaptation window closes, the client's progress becomes a direct reflection of whether the plan is actually building on itself, which is what the rest of this guide is about.

A USEFUL GUT CHECK

If you can't point to something that's objectively harder than it was four weeks ago (more weight, more reps, more sets, less rest), the client isn't on a program. They're just doing a workout, repeatedly.

The One Non-Negotiable

Progressive overload, gradually increasing the demand placed on the body over time, is about as close to settled science as exercise physiology gets. The underlying mechanism (rising mechanical tension driving the signalling behind strength and muscle growth) is basic, well-established physiology, not a contested theory.

Overload doesn't just mean adding weight

Adding load is the most obvious lever, but it's one of several. Any of the following count as genuine overload:

- ✓ More reps at the same weight
- ✓ More sets of the same exercise
- ✓ Less rest between sets (density)
- ✓ A harder variation of the same movement, or more range of motion
- ✓ The same weight for the same reps, with better technique or more control

Non-negotiable over a block, not every session

This is the nuance that gets lost: overload needs to be visible across weeks, not forced in every single workout. A beginner can genuinely adapt to a fixed stimulus for a while simply because it's new to them. Trying to add weight every session regardless of how the client is recovering is how good programming turns into burnout.

REAL EXAMPLE

A client has used the same 10kg dumbbells for eight weeks, working hard every time. Nothing has progressed, so nothing has overloaded, regardless of effort. The fix usually isn't heavier weight straight away, it's one more rep, or one more set, the following week.

Volume, Frequency & Intensity

These three variables are what you're actually adjusting every time you write a session. Getting the definitions straight makes every programming decision after this one easier.

- ✓ **Volume:** how much work is done, usually sets × reps, or total weight lifted (tonnage).
- ✓ **Intensity:** how heavy or hard it is, either as a percentage of a one-rep max, or as effort (RPE, or reps in reserve).
- ✓ **Frequency:** how often a muscle group or movement pattern is trained per week.

Volume landmarks: a useful mental model, not a formula

You may have come across MEV, MAV and MRV, the minimum, maximum-adaptive and maximum-recoverable volume for a muscle group in a week. The underlying research is real: more weekly sets are generally associated with more muscle growth, up to a point. But the specific set numbers attached to each landmark are a coaching heuristic built on top of that research, not independently validated fixed thresholds. They vary a lot between individuals and shift week to week with fatigue. Treat them as a mental model for reasoning about volume, not a formula to apply literally.

Frequency: the evidence-based floor

When total weekly volume is matched, current meta-analyses find training a muscle group twice a week produces meaningfully better results than once a week, and don't find a strong, consistent benefit to pushing frequency much higher than that on volume alone. In practice, twice a week per muscle group is a solid evidence-based floor. Beyond that, frequency becomes mostly a scheduling tool, a way to fit total weekly volume in without any single session becoming unmanageably long.

Intensity: effort matters more than the exact number

Chasing an exact percentage of a one-rep max is impractical for most PT clients, whose day-to-day strength and energy fluctuate. Reps in reserve (RIR), how many more reps a client could have done, is a simpler and more practical way to autoregulate load session to session. A client having an off day can hit the same RIR target with a lighter weight and still get a genuinely productive session.

EXAMPLE: REACHING A VOLUME TARGET WITHOUT OVERCOMPLICATING IT

A client doing 12 weekly sets of squats spread across two sessions (6 and 6) is easier to recover from and just as productive, evidence-wise, as trying to cram all 12 into one session, or spreading it across four separate visits they may not have time for.

Periodization Made Simple

Periodization just means planning how volume and intensity shift over time, rather than doing the same session indefinitely. Which model to use depends heavily on training experience and goal.

Linear periodization

Simple, good for beginners

- Start with higher reps and lighter load, gradually shift to lower reps and heavier load over a block.
- Easy to plan and explain, works well while a client is still making fast, straightforward progress.
- Less advantage once progress slows and a single fixed direction stops being enough stimulus.

Undulating (daily/weekly) periodization

Trained lifters chasing strength

- Rep ranges vary within the week, a heavy day, a moderate day, a lighter day, rather than shifting slowly over months.
- Meta-analyses show this outperforms linear periodization for strength gains, but specifically in already-trained lifters.
- Shows little to no advantage over linear for beginners, or for hypertrophy goals generally.

Block periodization

Performance and competition prep

- Dedicated phases (for example a hypertrophy block, then a strength block, then a peaking block).
- Suits competitive or performance-focused clients working toward a specific date.
- Usually more complexity than a general-population client actually needs.

DELOADS

A planned lower-intensity week every four to eight weeks is sound practical fatigue management, though it hasn't been isolated and directly proven the way progressive overload has. Periodized programs, which typically build in lighter phases, do outperform non-periodized ones for strength, which supports the logic even without a study testing deloads on their own.

One Size Doesn't Fit All

Everything so far is the floor: the principles that hold for almost everyone. What separates a good coach from a good template is reading the specific person in front of you, because two clients with an identical goal can need genuinely different programs to actually stick with it.

The routine-lovers

Some clients are motivated by visible, trackable progress on the same lifts, session after session. For them, keep the main lifts consistent for weeks at a time and let the numbers going up do the motivating. Changing exercises on this client for the sake of variety actively works against what keeps them engaged.

The variety-seekers

Other clients disengage quickly with repetition, often (though not only) general-population or health-motivated clients rather than competitive lifters. For them, rotate accessory exercises, tempos, or equipment regularly, while still anchoring one or two trackable movements per session so you can actually measure whether they're progressing underneath the variety.

Low-frequency clients need a different lens

A client training once or twice a week is one of the most common real-world cases where a textbook split simply doesn't work. A traditional body-part split (chest day, back day, leg day) means each muscle only gets trained once a week or less, which sits right at or below the evidence-based frequency floor from Chapter 3, and leaves almost no margin for a missed session. Run full-body sessions every time instead, keep one or two anchor movements (a squat pattern, a hinge pattern, a push, a pull) consistent for four to six weeks so there's something to actually track, and rotate everything else more freely than you would for a higher-frequency client, since variety matters more for adherence when there are simply fewer sessions for boredom to build up in.

THERE'S NO UNIVERSALLY "BETTER" PROGRAM

There's only the program a specific person will actually do, consistently, for months. A textbook-optimal plan a client abandons after three weeks has produced worse results than a slightly-less-optimal one they stuck with.

Programming for Men and Women

A few claims about sex differences in training circulate constantly in the fitness industry. Some hold up well under scrutiny. Others, including the popular idea that women can simply handle more training volume than men, are a real finding stretched further than the evidence actually supports. Both deserve an honest look.

What's genuinely well established

- ✓ **Absolute strength and power:** men typically produce greater absolute strength and power, largely due to greater muscle mass and more type II (fast-twitch) muscle fibre.
- ✓ **Relative strength is far more equal:** strength per unit of muscle cross-sectional area is essentially equivalent between sexes. A smaller, well-trained woman can be exactly as strong “for her size” as a bigger male client.
- ✓ **ACL injury risk:** women have roughly double the ACL injury rate of men in comparable sports, linked to landing mechanics (particularly increased inward knee movement on landing). This justifies deliberately programming landing technique and posterior-chain strength work, especially around jumping and change-of-direction movements.

The fatigue-resistance claim, more precisely

Women do show a genuine, well-replicated resistance to fatigue during sustained, submaximal, largely isometric efforts, that part is real. Where the popular version overreaches is in extending it to heavy, dynamic resistance training, and to next-day recovery: some research even finds the opposite there, with larger and longer-lasting strength and power decrements in women 24 to 72 hours after a hard session, in certain protocols. The honest summary is that the effect is task-specific, not a general rule that women fatigue less or recover faster across the board. Don't apply a blanket sex-based volume or frequency rule to any client: training history, recovery capacity and how someone actually responds week to week tell you far more than their sex does on its own.

ON THE MENSTRUAL CYCLE

Periodizing training around menstrual cycle phases is a popular idea, but current systematic reviews describe the evidence as inconsistent or inconclusive for anything like a rigid phase-based approach. It's a reasonable thing to discuss with a client who wants to track it for herself, but not a programming rule to apply by default.

From Framework to Real Session

None of the above matters if it doesn't turn into an actual session on an actual Tuesday. Here's how the pieces fit together, with real numbers.

A session has a shape

- ✓ **Warm-up:** general movement, then specific preparation for the day's main lift.
- ✓ **Primary lift:** the day's main trackable movement, where progressive overload is measured most closely.
- ✓ **Secondary and accessory work:** supports the primary lift, or targets a specific gap.
- ✓ **Optional finisher:** brief conditioning work, only if time and recovery allow.

Example: a 3x/week full-body block (beginner, general fitness)

Day A/B/C, rotating a squat, hinge, push and pull pattern across the week. Primary lifts run 3 sets of 8 to 10 reps at roughly RPE 7 (around 3 reps in reserve); accessories run 2 sets of 12 to 15 reps. Straightforward double progression: add a rep each session until the top of the range is hit on all sets, then add a small amount of load and drop back to the bottom of the range.

Example: a 1x/week client (low frequency)

Full body every session, same as the low-frequency approach in Chapter 5. The squat, hinge, push and pull anchors stay identical for four to six weeks so there's something consistent to track, while accessory exercise selection rotates weekly to keep the single weekly session feeling fresh.

WRITE IT DOWN, OR IT DIDN'T HAPPEN

A session your client (or you) can't recall next week is a session you can't actually overload. Track weight, reps and effort for every working set, it's the only way to know whether the plan on paper is actually producing the loop from Chapter One.

Common Programming Mistakes

Most stalled client progress traces back to one or two of these, quietly, over several weeks rather than one obvious failure.



Changing everything, every session

Novel exercises every time feels engaging but leaves nothing consistent enough to actually overload or measure.



Copy-pasting a program with no individualization

The same 12-week template handed to a routine-lover and a variety-seeker will fail one of them within a month.



Ignoring how the client actually feels that day

A rigid plan that never adjusts for sleep, stress or soreness produces worse sessions than one built around simple effort-based autoregulation.



Never deloading until an injury forces it

Planned lighter weeks are far cheaper than the unplanned week off that comes from ignoring accumulated fatigue.



Applying a sex-based rule instead of individualizing

Programming a client's volume off a generalization about men or women, rather than their own training history and recovery, ignores exactly the nuance covered in Chapter 6.



Chasing intensity over consistency for low-frequency clients

A brutal, unsustainable session once a week loses to a moderate one the client actually keeps showing up for.

The Programming Checklist

Every step from this guide, in order. Print this page or keep it open while you build your next block.

- ✓ Assess before you plan: goals, training history, injuries and constraints
- ✓ Confirm every block has a visible, trackable form of progressive overload
- ✓ Hit each muscle group or movement pattern at least twice a week where possible
- ✓ Use RPE or reps-in-reserve to autoregulate load on off days
- ✓ Pick a periodization model that matches training experience, not just preference
- ✓ Schedule a deload every four to eight weeks, before fatigue forces one
- ✓ Identify whether each client is a routine-lover or a variety-seeker, and program accordingly
- ✓ For 1–2x/week clients, run full-body sessions with consistent anchor lifts
- ✓ Program landing mechanics and posterior-chain strength for clients doing jumping or change-of-direction work
- ✓ Individualize volume and recovery by the client in front of you, not a sex-based rule of thumb
- ✓ Give every session a clear shape: warm-up, primary lift, accessories, optional finisher
- ✓ Use double progression (reps first, then load) as your default progression scheme
- ✓ Track every working set: weight, reps, and effort
- ✓ Review the last four weeks of data before writing the next block



A note from Totality's founder

"I built Totality because I'm a personal trainer myself, and I was sick of running my business through four apps that didn't talk to each other. This guide is the advice I wish someone had handed me when I started: the practical, unglamorous stuff nobody puts in a course brochure."

Taj Donaldson, Founder & CEO, Totality · Personal Trainer

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